Name:_____

Kinetic Energy

Energy is measured in _____, the symbol is ___.

Kinetic energy = the energy of m_____ is a form of energy so it is also measured in _____.

We can work out how much kinetic energy a moving object has by using the following formula

Kinetic energy= ½ X Objects mass X (objects speed)^2

$E_k = ½ ms^2$

Example. A girl has a mass of 60Kg. She is running at a speed of 2m/s. How much kinetic energy does she have?

Kinetic energy= ½ X Objects mass X (objects speed)^2

Kinetic energy= ½ X ____ kg X (____m/s)^2

E_k= ½ X 60 X _____

E_k = ½ X _____

E_k = _____ J

1. A man has a mass of 80Kg and is running at a speed of 2m/s. How much kinetic energy does he have?

Kinetic energy= ½ X Objects mass X (objects speed)^2

Kinetic energy= ½ X _____ X (_____)^2

E_k = ½ X _____ X _____

E_k =½ X _____

E_k = _____ *(don't forget the units)*

2. A car has a mass of 1000kg and is moving at 4m/s. How much kinetic energy does it have?

$E_k = ½ ms^2$

Kinetic energy= ½ X _____ X (_____)^2

E_k =

=

=

3. A ball has a mass of 0.5 kg and is moving at 4 m/s. What is its kinetic energy?

$E_k = ½ ms^2$

4. Two men are running at 2m/s. One man has a mass of 70kg and the other man has a mass of 80kg. Which man has the greatest kinetic energy? Show working for both men.

Measuring your kinetic energy!

You will be calculating the kinetic energy you have when you run across the basketball court.

Step one: measure the length of the basketball court using a trundle wheel.

LENGTH OF BASKETBALL COURT= _____m

Step two: measure the time it takes to run from one end of the basketball court to the other using a timer.

You need to work out your velocity in **metres per second**. Show your working below:

SPEED= _____

Step three: measure your mass using the scales.

MASS = _____

Step four: work out your kinetic energy in joules.

Kinetic energy = ½ x objects mass x (objects speed)^2 *or* $E_k = ½ ms^2$

Your kinetic energy in joules _____

Potential Energy

Potential energy is stored energy.

Food has potential energy stored in the _____ it is made up of.

A stretched elastic band has potential energy called _____ potential energy.

An object positioned above the ground has _____ potential energy.

Complete this table:

Situation	Type of potential energy(gravitational, elastic or chemical)
Wood used as fuel in a fire	
An apple in a tree	
A stretched spring	
Petrol as a fuel used in a car	

We can calculate the gravitational potential energy of an object.

Gravitational potential energy= Mass(kg) x 9.8 x Height (m)

$E_p = MGH$

Why does G = 9.8? _____

_____

What is the symbol for potential energy? _____

Example:

Bob has a mass of 90Kg. He is standing 2m above the ground. What is his gravitational potential energy?

$E_p = MGH$

E_P =_____ 90 x 9.8 x 2

M = _____ 1764j

G =_____ or 1.764kj

H = _____

Measuring your potential energy!

You will be calculating the potential energy you have when standing in a certain position.

Step one: find a location to stand and measure your distance off the ground.

Step two: measure your mass using the scales.

Step three: work out the gravity on earth.

Step four: work out your potential energy

Potential energy = objects mass x gravity x height *or* $E_p = mgh$

Your potential energy in joules _____

Step five: Repeat for a different position in the class.

E_P=

m=

g=

h=

Kinetic and Potential Energy Questions

1. A model glider has a mass of 1 kg. How much potential energy does it have 2 meters off the ground?

2. The same model has a velocity of 2.2 m/s. How much kinetic energy does it have?

3. A full-sized glider has a mass of 490kg, while it's pilot has a mass of 85kg. If it is 1,000 meters off the ground, how much potential energy do the plane and pilot have together?

4. The same glider from Problem 3 has a velocity of 35 m/s. How much kinetic energy does it have?

5. Kinetic energy can be defined as....

6. The equation to be used to calculate kinetic energy (E_k) is...

7. The SI unit in which E_k is often measured is the ..._____.

8. Rodger Maris swung a bat which had a mass of 2 Kg at a velocity of 45 m/s. How many joules of kinetic energy could he give to a ball?

9. Barry Bonds swings a bat which has a mass of 1.5 Kg at a velocity of 55 m/s. How many joules of kinetic energy could he give to a ball?

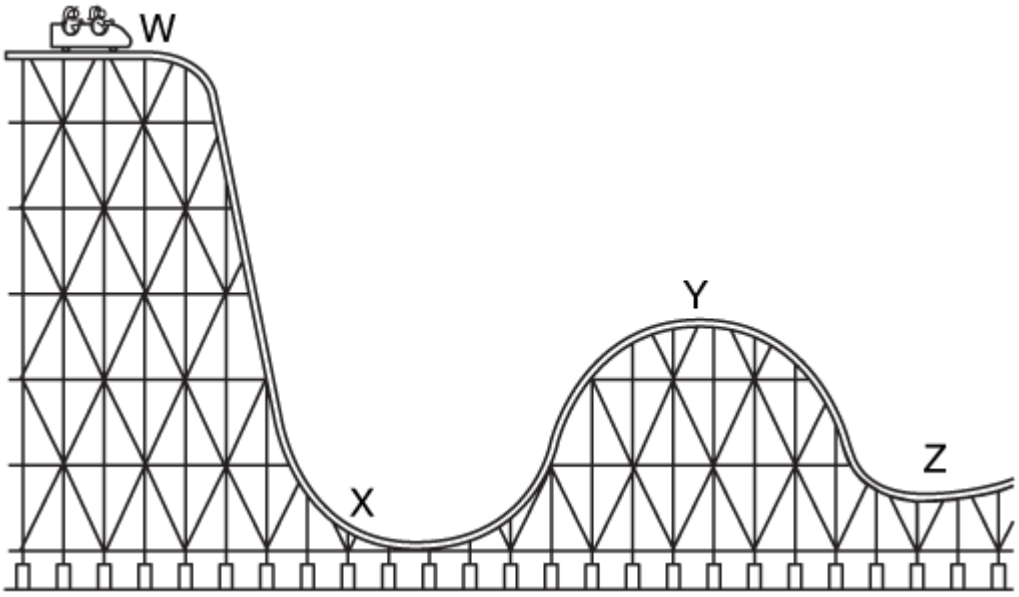
10. Which is more important to hitting a home run - a heavier bat or a faster swing?

11. A golf pro swings his driver which weighs .75 kg at a velocity of 60 m/s. Calculate the kinetic energy of the club.

12. Calculate the E_k of a car which has a mass of 1000 kg and is moving at the rate of 20 m/s.

13. What is the E_k of a soccer ball which has a mass of 0.8 kg and is kicked at a velocity of 10 m/s?

14. Calculate the E_k of a running back that has a mass of 80 kg and is running at a velocity of 8 m/s.



15. Describe the potential and kinetic energy the trolley would be experiencing at each point on the diagram.